

SpotBugs Report

Project Information

Project:

SpotBugs version: 4.7.3

Code analyzed:

- thingifier/target/classes/
- challenger/target/classes/

Metrics

8314 lines of code analyzed, in 167 classes, in 38 packages.

Metric	Total	Density*
High Priority Warnings	14	1.68
Medium Priority Warnings	155	18.64
Total Warnings	169	20.33

(* Defects per Thousand lines of non-commenting source statements)

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Summary

Warning Type	Number
Bad practice Warnings	7
Correctness Warnings	6
Internationalization Warnings	8
Malicious code vulnerability Warnings	101
Performance Warnings	27
Dodgy code Warnings	20
Total	169

Warnings

Click on a warning row to see full context information.

Bad practice Warnings

Code Warning

- DMI** Random object created and used only once in
uk.co.compendiumdev.challenge.practicemodes.simpleapi.SimpleAPITestDataPopulator.populate(ERSchema, ERInstanceData)
- DMI** Random object created and used only once in
uk.co.compendiumdev.thingifier.htmlgui.htmlgen.RestApiDocumentationGenerator.getExampleFilter(EntityDefinition)

- DMI** Random object created and used only once in
uk.co.compendiumdev.thingifier.htmlgui.htmlgen.RestApiDocumentationGenerator.getExampleFilter(EntityDefinition)
- OS** uk.co.compendiumdev.challenge.gui.MarkdownContentManager.getHtmlVersionOfMarkdownContent(String, String, Map) may fail to close stream
- OS** uk.co.compendiumdev.challenge.gui.MarkdownContentManager.getResourceAsString(String) may fail to close stream
- RV** Exceptional return value of java.io.File.mkdirs() ignored in
uk.co.compendiumdev.challenge.persistence.ChallengerFileStorage.saveChallengerStatus(ChallengerAuthData)
- RV** Exceptional return value of java.io.File.mkdirs() ignored in
uk.co.compendiumdev.challenge.persistence.ChallengerFileStorage.saveDatabaseContent(String, ERInstanceData)

Correctness Warnings

Code Warning

- NP** Possible null pointer dereference of challenger in
uk.co.compendiumdev.challenge.challengehooks.ChallengerInternalHTTPResponseHook.run(HttpApiRequest, InternalHttpResponse)
- NP** Possible null pointer dereference of relationshipToUse in uk.co.compendiumdev.thingifier.api.restapihandlers.RelationshipCreation.create(String, BodyParser, SimpleQuery, String) on exception path
- NP** Possible null pointer dereference of RestApiDocumentationGenerator.thingifier in
uk.co.compendiumdev.thingifier.htmlgui.htmlgen.RestApiDocumentationGenerator.getApiDocumentation(ApiRoutingDefinition, List, String, String)
- RCN** Nullcheck of challenge at line 653 of value previously dereferenced in
uk.co.compendiumdev.challenge.gui.ChallengerWebGUI.renderChallengeData(ChallengeDefinitions, ChallengerAuthData)
- RCN** Nullcheck of collection at line 27 of value previously dereferenced in
uk.co.compendiumdev.challenge.practicemodes.simpleapi.ResetAutoIncrementWhenTooHigh.run(HttpApiRequest, ThingifierApiConfig)
- RV** Return value of String.format(String, Object[]) ignored in
uk.co.compendiumdev.thingifier.application.MainImplementation.configureThingifierWithProfile()

Internationalization Warnings

Code Warning

- Dm** Found reliance on default encoding in
uk.co.compendiumdev.challenge.gui.MarkdownContentManager.getHtmlVersionOfMarkdownContent(String, String, Map): new java.io.InputStreamReader(InputStream)
- Dm** Found reliance on default encoding in uk.co.compendiumdev.challenge.gui.MarkdownContentManager.getResourceAsString(String): new java.io.InputStreamReader(InputStream)
- Dm** Found reliance on default encoding in uk.co.compendiumdev.challenge.persistence.AwsS3Storage.getObjectContent(InputStream): new java.io.InputStreamReader(InputStream)
- Dm** Found reliance on default encoding in uk.co.compendiumdev.challenge.persistence.ChallengerFileStorage.loadChallengerStatus(String): new String(byte[])
- Dm** Found reliance on default encoding in uk.co.compendiumdev.challenge.persistence.ChallengerFileStorage.loadDatabaseContent(String): new String(byte[])
- Dm** Found reliance on default encoding in
uk.co.compendiumdev.challenge.persistence.ChallengerFileStorage.saveChallengerStatus(ChallengerAuthData): String.getBytes()
- Dm** Found reliance on default encoding in uk.co.compendiumdev.challenge.persistence.ChallengerFileStorage.saveDatabaseContent(String, ERInstanceData): String.getBytes()
- Dm** Found reliance on default encoding in uk.co.compendiumdev.thingifier.api.http.headers.headerparser.BasicAuthHeaderParser.decodeBase64(): new String(byte[])

Malicious code vulnerability Warnings

Code Warning

- EI** uk.co.compendiumdev.challenge.ChallengeRouteHandler.getChallengers() may expose internal representation by returning ChallengeRouteHandler.challengers
- EI** uk.co.compendiumdev.challenge.ChallengeRouteHandler.getThingifier() may expose internal representation by returning ChallengeRouteHandler.thingifier
- EI** uk.co.compendiumdev.challenge.challengers.Challengers.getApiConfig() may expose internal representation by returning Challengers.apiConfig
- EI** uk.co.compendiumdev.challenge.challengers.Challengers.getDefinedChallenges() may expose internal representation by returning Challengers.definedChallenges
- EI** uk.co.compendiumdev.challenge.challenges.ChallengeDefinitions.getChallengeSections() may expose internal representation by returning ChallengeDefinitions.sections
- EI** uk.co.compendiumdev.challenge.challenges.ChallengeDefinitions.getChallenges() may expose internal representation by returning ChallengeDefinitions.orderedChallenges
- EI** uk.co.compendiumdev.challenge.challenges.ChallengeSection.getChallenges() may expose internal representation by returning ChallengeSection.challenges
- EI** uk.co.compendiumdev.challenge.persistence.PersistenceResponse.getAuthData() may expose internal representation by returning PersistenceResponse.authData
- EI** uk.co.compendiumdev.thingifier.Thingifier.apiConfig() may expose internal representation by returning Thingifier.apiConfig
- EI** uk.co.compendiumdev.thingifier.api.docgen.ApiRoutingDefinition.definitions() may expose internal representation by returning ApiRoutingDefinition.routing

- EI** uk.co.compendiumdev.thingifier.api.docgen.RoutingDefinition.getPossibleStatusResponses() may expose internal representation by returning RoutingDefinition.possibleStatusResponses
- EI** uk.co.compendiumdev.thingifier.api.docgen.ThingifierApiDocumentationDefn.getAdditionalRoutes() may expose internal representation by returning ThingifierApiDocumentationDefn.additionalRoutes
- EI** uk.co.compendiumdev.thingifier.api.docgen.ThingifierApiDocumentationDefn.getServers() may expose internal representation by returning ThingifierApiDocumentationDefn.servers
- EI** uk.co.compendiumdev.thingifier.api.docgen.ThingifierApiDocumentationDefn.getThingifier() may expose internal representation by returning ThingifierApiDocumentationDefn.thingifier
- EI** uk.co.compendiumdev.thingifier.api.http.HttpApiRequest.getHeaders() may expose internal representation by returning HttpApiRequest.headers
- EI** uk.co.compendiumdev.thingifier.api.http.HttpApiRequest.getQueryParams() may expose internal representation by returning HttpApiRequest.queryParams
- EI** uk.co.compendiumdev.thingifier.api.http.HttpApiRequestValidator.getErrorApiResponse() may expose internal representation by returning HttpApiRequestValidator.errorResponse
- EI** uk.co.compendiumdev.thingifier.api.http.HttpApiResponse.apiResponse() may expose internal representation by returning HttpApiResponse.apiResponse
- EI** uk.co.compendiumdev.thingifier.api.http.HttpApiResponse.getHeaders() may expose internal representation by returning HttpApiResponse.apiResponseHeaders
- EI** uk.co.compendiumdev.thingifier.api.http.bodyparser.BodyParser.getMap() may expose internal representation by returning BodyParser.args
- EI** uk.co.compendiumdev.thingifier.api.response.ApiResponse.getErrorMessages() may expose internal representation by returning ApiResponse.errorMessages
- EI** uk.co.compendiumdev.thingifier.api.response.ApiResponse.getHeaders() may expose internal representation by returning ApiResponse.headers
- EI** uk.co.compendiumdev.thingifier.api.response.ApiResponse.getReturnedInstanceCollection() may expose internal representation by returning ApiResponse.thingsToReturn
- EI** uk.co.compendiumdev.thingifier.api.restapihandlers.RelationshipCollector.getRelationshipDetails() may expose internal representation by returning RelationshipCollector.theRelationshipDetails
- EI** uk.co.compendiumdev.thingifier.api.restapihandlers.RelationshipCollector.getRelationshipsKeys() may expose internal representation by returning RelationshipCollector.identifiedRelationships
- EI** uk.co.compendiumdev.thingifier.apiconfig.ThingifierApiConfig.adminConfig() may expose internal representation by returning ThingifierApiConfig.adminConfig
- EI** uk.co.compendiumdev.thingifier.apiconfig.ThingifierApiConfig.forParams() may expose internal representation by returning ThingifierApiConfig.paramsConfig
- EI** uk.co.compendiumdev.thingifier.apiconfig.ThingifierApiConfig.jsonOutput() may expose internal representation by returning ThingifierApiConfig.jsonOutputConfig
- EI** uk.co.compendiumdev.thingifier.apiconfig.ThingifierApiConfig.statusCodes() may expose internal representation by returning ThingifierApiConfig.statusCodeConfig
- EI** uk.co.compendiumdev.thingifier.apiconfig.ThingifierApiConfigProfile.apiConfig() may expose internal representation by returning ThingifierApiConfigProfile.apiConfig
- EI** uk.co.compendiumdev.thingifier.application.AdhocDocumentedSparkRouteConfigurer.getApiDocDefn() may expose internal representation by returning AdhocDocumentedSparkRouteConfigurer.apiDefn
- EI** uk.co.compendiumdev.thingifier.application.MainImplementation.chooseThingifier(String) may expose internal representation by returning MainImplementation.thingifier
- EI** uk.co.compendiumdev.thingifier.application.MainImplementation.getApiDefn() may expose internal representation by returning MainImplementation.apiDefn
- EI** uk.co.compendiumdev.thingifier.application.MainImplementation.getGuiManagement() may expose internal representation by returning MainImplementation.guiManagement
- EI** uk.co.compendiumdev.thingifier.application.internalhttpconversion.InternalHttpResponse.getHeaders() may expose internal representation by returning InternalHttpResponse.headers
- EI** uk.co.compendiumdev.thingifier.application.routehandlers.ShutdownRouteHandler.getRoutes() may expose internal representation by returning ShutdownRouteHandler.routes
- EI2** new uk.co.compendiumdev.challenge.ChallengeRouteHandler(Thingifier, ThingifierApiDocumentationDefn, ChallengerConfig) may expose internal representation by storing an externally mutable object into ChallengeRouteHandler.apiChallengesDocumentationDefn
- EI2** new uk.co.compendiumdev.challenge.ChallengeRouteHandler(Thingifier, ThingifierApiDocumentationDefn, ChallengerConfig) may expose internal representation by storing an externally mutable object into ChallengeRouteHandler.thingifier
- EI2** uk.co.compendiumdev.challenge.ChallengeRouteHandler.setupGui(DefaultGUIHTML) may expose internal representation by storing an externally mutable object into ChallengeRouteHandler.guiTemplates
- EI2** new uk.co.compendiumdev.challenge.challengehooks.ChallengerApiRequestHook(Challengers) may expose internal representation by storing an externally mutable object into ChallengerApiRequestHook.challengers
- EI2** new uk.co.compendiumdev.challenge.challengehooks.ChallengerApiResponseHook(Challengers, Thingifier) may expose internal representation by storing an externally mutable object into ChallengerApiResponseHook.challengers
- EI2** new uk.co.compendiumdev.challenge.challengehooks.ChallengerApiResponseHook(Challengers, Thingifier) may expose internal representation by storing an externally mutable object into ChallengerApiResponseHook.thingifier
- EI2** new uk.co.compendiumdev.challenge.challengehooks.ChallengerInternalHTTPRequestHook(Challengers) may expose internal representation by storing an externally mutable object into ChallengerInternalHTTPRequestHook.challengers
- EI2** new uk.co.compendiumdev.challenge.challengehooks.ChallengerInternalHTTPResponseHook(Challengers) may expose internal representation by storing an externally mutable object into ChallengerInternalHTTPResponseHook.challengers
- EI2** new uk.co.compendiumdev.challenge.challengehooks.Challengers(EntityRelModel, Collection) may expose internal representation by storing an externally mutable object into Challengers.definedChallenges
- EI2** uk.co.compendiumdev.challenge.challengers.Challengers.setApiConfig(ThingifierApiConfig) may expose internal representation by storing an externally mutable object into Challengers.apiConfig
- EI2** new uk.co.compendiumdev.challenge.gui.ChallengerWebGUI(DefaultGUIHTML, boolean) may expose internal representation by storing an externally mutable object into ChallengerWebGUI.guiManagement

- EI2** new uk.co.compendiumdev.challenge.gui.MarkdownContentManager(List, DefaultGUIHTML) may expose internal representation by storing an externally mutable object into MarkdownContentManager.guiManagement
- EI2** new uk.co.compendiumdev.challenge.gui.PageNotFoundResponse(DefaultGUIHTML) may expose internal representation by storing an externally mutable object into PageNotFoundResponse.guiManagement
- EI2** uk.co.compendiumdev.challenge.persistence.PersistenceResponse.withChallengerAuthData(ChallengerAuthData) may expose internal representation by storing an externally mutable object into PersistenceResponse.authData
- EI2** new uk.co.compendiumdev.challenge.practicemodes.simpleapi.SimpleApiRoutes(DefaultGUIHTML) may expose internal representation by storing an externally mutable object into SimpleApiRoutes.guiTemplates
- EI2** new uk.co.compendiumdev.challenge.practicemodes.simulation.SimulationRoutes(DefaultGUIHTML) may expose internal representation by storing an externally mutable object into SimulationRoutes.guiTemplates
- EI2** new uk.co.compendiumdev.thingifier.Thingifier(EntityRelModel, ThingifierApiConfig, ThingifierApiConfigProfiles, String, String, ApiDocsConfig) may expose internal representation by storing an externally mutable object into Thingifier.apiConfig
- EI2** new uk.co.compendiumdev.thingifier.api.ApiUrls(ThingifierApiConfig) may expose internal representation by storing an externally mutable object into ApiUrls.apiConfig
- EI2** new uk.co.compendiumdev.thingifier.api.docgen.ApiRoutingDefinitionDocGenerator(Thingifier) may expose internal representation by storing an externally mutable object into ApiRoutingDefinitionDocGenerator.thingifier
- EI2** uk.co.compendiumdev.thingifier.api.docgen.ThingifierApiDocumentationDefn.setThingifier(Thingifier) may expose internal representation by storing an externally mutable object into ThingifierApiDocumentationDefn.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.docgen.ThingifierApiDocumentationDefn\$ApiServer(ThingifierApiDocumentationDefn, String, String) may expose internal representation by storing an externally mutable object into ThingifierApiDocumentationDefn\$ApiServer.this\$0
- EI2** new uk.co.compendiumdev.thingifier.api.ermodeconversion.JsonThing(JsonOutputConfig) may expose internal representation by storing an externally mutable object into JsonThing.apiConfig
- EI2** uk.co.compendiumdev.thingifier.api.http.HttpApiRequest.setQueryParams(Map) may expose internal representation by storing an externally mutable object into HttpApiRequest.queryParams
- EI2** uk.co.compendiumdev.thingifier.api.http.HttpApiRequest.setRawQueryParams(Map) may expose internal representation by storing an externally mutable object into HttpApiRequest.rawQueryParams
- EI2** new uk.co.compendiumdev.thingifier.api.http.HttpApiRequestValidator(ThingifierApiConfig) may expose internal representation by storing an externally mutable object into HttpApiRequestValidator.apiConfig
- EI2** new uk.co.compendiumdev.thingifier.api.http.HttpApiResponse(HttpHeadersBlock, ApiResponse, JsonThing, ThingifierApiConfig) may expose internal representation by storing an externally mutable object into HttpApiResponse.apiConfig
- EI2** new uk.co.compendiumdev.thingifier.api.http.HttpApiResponse(HttpHeadersBlock, ApiResponse, JsonThing, ThingifierApiConfig) may expose internal representation by storing an externally mutable object into HttpApiResponse.apiResponse
- EI2** new uk.co.compendiumdev.thingifier.api.http.ThingifierHttpApi(Thingifier, List, List) may expose internal representation by storing an externally mutable object into ThingifierHttpApi.apiRequestHooks
- EI2** new uk.co.compendiumdev.thingifier.api.http.ThingifierHttpApi(Thingifier, List, List) may expose internal representation by storing an externally mutable object into ThingifierHttpApi.apiResponseHooks
- EI2** new uk.co.compendiumdev.thingifier.api.http.ThingifierHttpApi(Thingifier, List, List) may expose internal representation by storing an externally mutable object into ThingifierHttpApi.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.http.bodyparser.BodyParser(HttpApiRequest, List) may expose internal representation by storing an externally mutable object into BodyParser.request
- EI2** new uk.co.compendiumdev.thingifier.api.http.bodyparser.BodyParser(HttpApiRequest, List) may expose internal representation by storing an externally mutable object into BodyParser.thingNames
- EI2** new uk.co.compendiumdev.thingifier.api.http.bodyparser.xml.XMLParserUsingOrgJson(String, List) may expose internal representation by storing an externally mutable object into XMLParserUsingOrgJson.thingNames
- EI2** new uk.co.compendiumdev.thingifier.api.http.headers.headervalidator.AcceptHeaderValidator(ThingifierApiConfig) may expose internal representation by storing an externally mutable object into AcceptHeaderValidator.apiConfig
- EI2** new uk.co.compendiumdev.thingifier.api.http.headers.headervalidator.ContentTypeHeaderValidator(ThingifierApiConfig) may expose internal representation by storing an externally mutable object into ContentTypeHeaderValidator.apiConfig
- EI2** new uk.co.compendiumdev.thingifier.api.response.ApiResponseAsJson(ApiResponse, JsonThing) may expose internal representation by storing an externally mutable object into ApiResponseAsJson.apiResponse
- EI2** new uk.co.compendiumdev.thingifier.api.response.ApiResponseAsXml(ApiResponse, JsonThing) may expose internal representation by storing an externally mutable object into ApiResponseAsXml.apiResponse
- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.BodyArgsProcessor(Thingifier, BodyParser) may expose internal representation by storing an externally mutable object into BodyArgsProcessor.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.BodyCreationValidator(Thingifier) may expose internal representation by storing an externally mutable object into BodyCreationValidator.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.BodyRelationshipValidator(Thingifier) may expose internal representation by storing an externally mutable object into BodyRelationshipValidator.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.RelationshipCreation(Thingifier) may expose internal representation by storing an externally mutable object into RelationshipCreation.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.RelationshipCreator(Thingifier) may expose internal representation by storing an externally mutable object into RelationshipCreator.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.RestApiDeleteHandler(Thingifier) may expose internal representation by storing an externally mutable object into RestApiDeleteHandler.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.RestApiGetHandler(Thingifier) may expose internal representation by storing an externally mutable object into RestApiGetHandler.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.RestApiPostHandler(Thingifier) may expose internal representation by storing an externally mutable object into RestApiPostHandler.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.RestApiPutHandler(Thingifier) may expose internal representation by storing an externally mutable object into RestApiPutHandler.thingifier
- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.ThingAmendment(Thingifier) may expose internal representation by storing an externally mutable object into ThingAmendment.thingifier

- EI2** new uk.co.compendiumdev.thingifier.api.restapihandlers.ThingCreation(Thingifier) may expose internal representation by storing an externally mutable object into ThingCreation.thingifier
- EI2** new uk.co.compendiumdev.thingifier.application.AdhocDocumentedSparkRouteConfigurer(ThingifierApiDocumentationDefn) may expose internal representation by storing an externally mutable object into AdhocDocumentedSparkRouteConfigurer.apiDefn
- EI2** uk.co.compendiumdev.thingifier.application.MainImplementation.setDefaultsFromArgs(String[]) may expose internal representation by storing an externally mutable object into MainImplementation.args
- EI2** new uk.co.compendiumdev.thingifier.application.httprouting.ThingifierHttpApiRoutings(Thingifier, ThingifierApiDocumentationDefn) may expose internal representation by storing an externally mutable object into ThingifierHttpApiRoutings.apiDefn
- EI2** new uk.co.compendiumdev.thingifier.application.internalhttpconversion.ThingifierHttpApiBridge(Thingifier, List, List) may expose internal representation by storing an externally mutable object into ThingifierHttpApiBridge.apiRequestHooks
- EI2** new uk.co.compendiumdev.thingifier.application.internalhttpconversion.ThingifierHttpApiBridge(Thingifier, List, List) may expose internal representation by storing an externally mutable object into ThingifierHttpApiBridge.apiResponseHooks
- EI2** new uk.co.compendiumdev.thingifier.application.internalhttpconversion.ThingifierHttpApiBridge(Thingifier, List, List) may expose internal representation by storing an externally mutable object into ThingifierHttpApiBridge.thingifier
- EI2** new uk.co.compendiumdev.thingifier.application.routehandlers.SparkApiRequestResponseHandler(Request, Response, Thingifier) may expose internal representation by storing an externally mutable object into SparkApiRequestResponseHandler.thingifier
- EI2** new uk.co.compendiumdev.thingifier.application.sparkhttpmessageHooks.ClearDataPreSparkRequestHook(int, Thingifier) may expose internal representation by storing an externally mutable object into ClearDataPreSparkRequestHook.thingifier
- EI2** new uk.co.compendiumdev.thingifier.htmlgui.htmlgen.DefaultGuiHtmlPages(DefaultGUIHTML, Thingifier, String) may expose internal representation by storing an externally mutable object into DefaultGuiHtmlPages.templates
- EI2** new uk.co.compendiumdev.thingifier.htmlgui.htmlgen.DefaultGuiHtmlPages(DefaultGUIHTML, Thingifier, String) may expose internal representation by storing an externally mutable object into DefaultGuiHtmlPages.thingifier
- EI2** new uk.co.compendiumdev.thingifier.htmlgui.htmlgen.RestApiDocumentationGenerator(Thingifier, DefaultGUIHTML) may expose internal representation by storing an externally mutable object into RestApiDocumentationGenerator.defaultGui
- EI2** new uk.co.compendiumdev.thingifier.htmlgui.htmlgen.RestApiDocumentationGenerator(Thingifier, DefaultGUIHTML) may expose internal representation by storing an externally mutable object into RestApiDocumentationGenerator.thingifier
- EI2** new uk.co.compendiumdev.thingifier.htmlgui.routing.DefaultGuiRoutings(Thingifier, DefaultGUIHTML) may expose internal representation by storing an externally mutable object into DefaultGuiRoutings.templates
- EI2** new uk.co.compendiumdev.thingifier.htmlgui.routing.DefaultGuiRoutings(Thingifier, DefaultGUIHTML) may expose internal representation by storing an externally mutable object into DefaultGuiRoutings.thingifier
- EI2** new uk.co.compendiumdev.thingifier.reporting.ThingReporter(Thingifier) may expose internal representation by storing an externally mutable object into ThingReporter.thingifier
- EI2** new uk.co.compendiumdev.thingifier.swaggerizer.Swaggerizer(ThingifierApiDocumentationDefn) may expose internal representation by storing an externally mutable object into Swaggerizer.apiDefn
- MS** Public static uk.co.compendiumdev.challenge.ChallengeMain.getChallenger() may expose internal representation by returning ChallengeMain.challenger

Performance Warnings

Code Warning

- SBSC** uk.co.compendiumdev.challenge.persistence.AwsS3Storage.getObjectContent(InputStream) concatenates strings using + in a loop
- SBSC** uk.co.compendiumdev.thingifier.api.restapihandlers.RelationshipCreation.create(String, BodyParser, SimpleQuery, String) concatenates strings using + in a loop
- SBSC** uk.co.compendiumdev.thingifier.api.restapihandlers.RestApiPutHandler.handle(String, BodyParser, HttpHeadersBlock) concatenates strings using + in a loop
- SIC** Should uk.co.compendiumdev.thingifier.api.docgen.ThingifierApiDocumentationDefn\$ApiServer be a _static_ inner class?
- SIC** Should uk.co.compendiumdev.thingifier.htmlgui.htmlgen.DefaultGUIHTML\$GuiMenuItem be a _static_ inner class?
- SS** Unread field: uk.co.compendiumdev.challenge.challengesrouting.ChallengerTrackingRoutes.MAX_CHALLENGERS_PER_IP; should this field be static?
- SS** Unread field: uk.co.compendiumdev.thingifier.api.docgen.ApiRoutingDefinitionDocGenerator.uniqueGUID; should this field be static?
- SS** Unread field: uk.co.compendiumdev.thingifier.api.docgen.ApiRoutingDefinitionDocGenerator.uniqueID; should this field be static?
- UrF** Unread field: uk.co.compendiumdev.challenge.practicemodes.simpleapi.SimpleApiRoutes.simpleApiDocsRouting
- UrF** Unread field: uk.co.compendiumdev.challenge.practicemodes.simpleapi.SimpleApiRoutes.simpleApiGuiRouting
- UrF** Unread field: uk.co.compendiumdev.challenge.practicemodes.simulation.SimulationRoutes.httpApi
- UrF** Unread field: uk.co.compendiumdev.challenge.practicemodes.simulation.SimulationRoutes.jsonThing
- UrF** Unread field: uk.co.compendiumdev.challenge.practicemodes.simulation.SimulationRoutes.simulatorDocsRouting
- UrF** Unread field: uk.co.compendiumdev.thingifier.apiconfig.ApiDocsConfig.apiIntroductionParaOverride
- UrF** Unread field: uk.co.compendiumdev.thingifier.apiconfig.StatusCodeConfig.maxRequestContentLengthBytes
- UrF** Unread field: uk.co.compendiumdev.thingifier.application.MainImplementation.args
- UrF** Unread field: uk.co.compendiumdev.thingifier.application.MainImplementation.docsServerRouting
- UrF** Unread field: uk.co.compendiumdev.thingifier.application.MainImplementation.exceptionRoutings
- UrF** Unread field: uk.co.compendiumdev.thingifier.application.internalhttpconversion.ThingifierHttpApiBridge.apiRequestHooks
- UrF** Unread field: uk.co.compendiumdev.thingifier.application.internalhttpconversion.ThingifierHttpApiBridge.apiResponseHooks
- WMI** uk.co.compendiumdev.challenge.challengesrouting.ChallengerIpAddressTracker.purgeEmptyIpAddresses(Set) makes inefficient use of keySet iterator instead of entrySet iterator
- WMI** uk.co.compendiumdev.challenge.gui.ChallengerWebGUI.setup(Challengers, ChallengeDefinitions, PersistenceLayer, boolean) makes inefficient use of keySet iterator instead of entrySet iterator
- WMI** uk.co.compendiumdev.challenge.gui.MarkdownContentManager.processMacrosInContentLine(String, Map) makes inefficient use of keySet iterator instead of entrySet iterator

- WMI** uk.co.compendiumdev.thingifier.api.http.bodyparser.BodyParser.getObjectNames() makes inefficient use of keySet iterator instead of entrySet iterator
- WMI** uk.co.compendiumdev.thingifier.api.http.bodyparser.BodyParser.stringMap(Map) makes inefficient use of keySet iterator instead of entrySet iterator
- WMI** uk.co.compendiumdev.thingifier.api.restapihandlers.RelationshipCreation.create(String, BodyParser, SimpleQuery, String) makes inefficient use of keySet iterator instead of entrySet iterator
- WMI** uk.co.compendiumdev.thingifier.htmlgui.htmlgen.DefaultGuiHtmlPages.getInstanceDetailsPage(String, String, Map) makes inefficient use of keySet iterator instead of entrySet iterator

Dodgy code Warnings

Code Warning

- DLS** Dead store to todoId in uk.co.compendiumdev.challenge.challengehooks.ChallengerApiResponseHook.run(HttpApiRequest, ApiResponse, ThingifierApiConfig)
- DLS** Dead store to bcHeader in uk.co.compendiumdev.challenge.gui.MarkdownContentManager.getHtmlVersionOfMarkdownContent(String, String, Map)
- DLS** Dead store to mirrorDocsRouting in uk.co.compendiumdev.challenge.practicemodes.mirror.MirrorRoutes.configure(ThingifierApiDocumentationDefn, DefaultGUIHTML)
- DLS** Dead store to relationshipsPart in uk.co.compendiumdev.thingifier.api.restapihandlers.BodyRelationshipValidator.validateComplexFourPartRelationshipDefinition(EntityDefinition, ValidationReport, String, String, String)
- DLS** Dead store to args in uk.co.compendiumdev.thingifier.api.restapihandlers.ThingAmendment.amendInstance(BodyParser, EntityInstance, Boolean, String)
- DLS** Dead store to relresponse in uk.co.compendiumdev.thingifier.api.restapihandlers.ThingAmendment.amendInstance(BodyParser, EntityInstance, Boolean, String)
- ICAST** Result of integer multiplication cast to long in new uk.co.compendiumdev.thingifier.application.sparkhttpmessagehooks.ClearDataPreSparkRequestHook(int, Thingifier)
- NP** Load of known null value in uk.co.compendiumdev.challenge.challengehooks.ChallengerInternalHttpResponseHook.run(HttpApiRequest, InternalHttpResponse)
- NP** Load of known null value in uk.co.compendiumdev.challenge.challengesrouting.AuthRoutes.lambda\$configure\$1(Challengers, Request, Response)
- NP** Load of known null value in uk.co.compendiumdev.challenge.challengesrouting.AuthRoutes.lambda\$configure\$3(Challengers, Request, Response)
- NP** Load of known null value in uk.co.compendiumdev.challenge.challengesrouting.AuthRoutes.lambda\$configure\$6(Challengers, Request, Response)
- RCN** Redundant nullcheck of challenger, which is known to be non-null in uk.co.compendiumdev.challenge.challengehooks.ChallengerInternalHttpResponseHook.run(HttpApiRequest, InternalHttpResponse)
- RV** Return value of uk.co.compendiumdev.thingifier.apiconfig.ParamConfig.willAllowFilteringThroughUrlParams() ignored, but method has no side effect
- RV** Return value of uk.co.compendiumdev.thingifier.apiconfig.ParamConfig.willEnforceFilteringThroughUrlParams() ignored, but method has no side effect
- SF** Switch statement found in uk.co.compendiumdev.thingifier.api.http.ThingifierHttpApi.routeAndProcessRequest(HttpApiRequest, ThingifierHttpApi\$HttpVerb) where default case is missing
- SF** Switch statement found in uk.co.compendiumdev.thingifier.spark.SimpleSparkRouteCreator.addHandler(String, String, Route) where default case is missing
- SF** Switch statement found in uk.co.compendiumdev.thingifier.swaggerizer.Swaggerizer.setOperationVerb(RoutingDefinition, PathItem, Operation) where default case is missing
- ST** Write to static field uk.co.compendiumdev.challenge.persistence.PersistenceLayer.aws from instance method new uk.co.compendiumdev.challenge.persistence.PersistenceLayer(PersistenceLayer\$StorageType)
- UC** Useless object stored in variable bcHeader of method uk.co.compendiumdev.challenge.gui.MarkdownContentManager.getHtmlVersionOfMarkdownContent(String, String, Map)
- UrF** Unread public/protected field: uk.co.compendiumdev.challenge.challenges.ChallengeData.status

Details

DLS_DEAD_LOCAL_STORE: Dead store to local variable

This instruction assigns a value to a local variable, but the value is not read or used in any subsequent instruction. Often, this indicates an error, because the value computed is never used.

Note that Sun's javac compiler often generates dead stores for final local variables. Because SpotBugs is a bytecode-based tool, there is no easy way to eliminate these false positives.

DMI_RANDOM_USED_ONLY_ONCE: Random object created and used only once

This code creates a java.util.Random object, uses it to generate one random number, and then discards the Random object. This produces mediocre quality random numbers and is inefficient. If possible, rewrite the code so that the Random object is created once and saved, and each time a new random number is required invoke a method on the existing Random object to obtain it.

If it is important that the generated Random numbers not be guessable, you *must* not create a new Random for each random number; the values are too easily guessable. You should strongly consider using a java.security.SecureRandom instead (and avoid allocating a new SecureRandom for each random

number needed).

DM_DEFAULT_ENCODING: Reliance on default encoding

Found a call to a method which will perform a byte to String (or String to byte) conversion, and will assume that the default platform encoding is suitable. This will cause the application behavior to vary between platforms. Use an alternative API and specify a charset name or Charset object explicitly.

EI_EXPOSE_REP: May expose internal representation by returning reference to mutable object

Returning a reference to a mutable object value stored in one of the object's fields exposes the internal representation of the object. If instances are accessed by untrusted code, and unchecked changes to the mutable object would compromise security or other important properties, you will need to do something different. Returning a new copy of the object is better approach in many situations.

EI_EXPOSE_REP2: May expose internal representation by incorporating reference to mutable object

This code stores a reference to an externally mutable object into the internal representation of the object. If instances are accessed by untrusted code, and unchecked changes to the mutable object would compromise security or other important properties, you will need to do something different. Storing a copy of the object is better approach in many situations.

ICAST_INTEGER_MULTIPLY_CAST_TO_LONG: Result of integer multiplication cast to long

This code performs integer multiply and then converts the result to a long, as in:

```
long convertDaysToMilliseconds(int days) { return 1000*3600*24*days; }
```

If the multiplication is done using long arithmetic, you can avoid the possibility that the result will overflow. For example, you could fix the above code to:

```
long convertDaysToMilliseconds(int days) { return 1000L*3600*24*days; }
```

or

```
static final long MILLISECONDS_PER_DAY = 24L*3600*1000;  
long convertDaysToMilliseconds(int days) { return days * MILLISECONDS_PER_DAY; }
```

MS_EXPOSE_REP: Public static method may expose internal representation by returning array

A public static method returns a reference to an array that is part of the static state of the class. Any code that calls this method can freely modify the underlying array. One fix is to return a copy of the array.

NP_LOAD_OF_KNOWN_NULL_VALUE: Load of known null value

The variable referenced at this point is known to be null due to an earlier check against null. Although this is valid, it might be a mistake (perhaps you intended to refer to a different variable, or perhaps the earlier check to see if the variable is null should have been a check to see if it was non-null).

NP_NULL_ON_SOME_PATH: Possible null pointer dereference

There is a branch of statement that, *if executed*, guarantees that a null value will be dereferenced, which would generate a `NullPointerException` when the code is executed. Of course, the problem might be that the branch or statement is infeasible and that the null pointer exception cannot ever be executed; deciding that is beyond the ability of SpotBugs.

NP_NULL_ON_SOME_PATH_EXCEPTION: Possible null pointer dereference in method on exception path

A reference value which is null on some exception control path is dereferenced here. This may lead to a `NullPointerException` when the code is executed. Note that because SpotBugs currently does not prune infeasible exception paths, this may be a false warning.

Also note that SpotBugs considers the default case of a switch statement to be an exception path, since the default case is often infeasible.

OS_OPEN_STREAM: Method may fail to close stream

The method creates an IO stream object, does not assign it to any fields, pass it to other methods that might close it, or return it, and does not appear to close the stream on all paths out of the method. This may result in a file descriptor leak. It is generally a good idea to use a `finally` block to ensure that streams are closed.

RCN_REDUNDANT_NULLCHECK_WOULD_HAVE_BEEN_A_NPE: Nullcheck of value previously dereferenced

A value is checked here to see whether it is null, but this value cannot be null because it was previously dereferenced and if it were null a null pointer exception would have occurred at the earlier dereference. Essentially, this code and the previous dereference disagree as to whether this value is allowed to be null. Either the check is redundant or the previous dereference is erroneous.

RCN_REDUNDANT_NULLCHECK_OF_NONNULL_VALUE: Redundant nullcheck of value known to be non-null

This method contains a redundant check of a known non-null value against the constant null.

RV_RETURN_VALUE_IGNORED_BAD_PRACTICE: Method ignores exceptional return value

This method returns a value that is not checked. The return value should be checked since it can indicate an unusual or unexpected function execution. For example, the `File.delete()` method returns false if the file could not be successfully deleted (rather than throwing an Exception). If you don't check the result, you won't notice if the method invocation signals unexpected behavior by returning an atypical return value.

RV_RETURN_VALUE_IGNORED: Method ignores return value

The return value of this method should be checked. One common cause of this warning is to invoke a method on an immutable object, thinking that it updates the object. For example, in the following code fragment,

```
String dateString = getHeaderField(name);
dateString.trim();
```

the programmer seems to be thinking that the `trim()` method will update the String referenced by `dateString`. But since Strings are immutable, the `trim()` function returns a new String value, which is being ignored here. The code should be corrected to:

```
String dateString = getHeaderField(name);
dateString = dateString.trim();
```

RV_RETURN_VALUE_IGNORED_NO_SIDE_EFFECT: Return value of method without side effect is ignored

This code calls a method and ignores the return value. However our analysis shows that the method (including its implementations in subclasses if any) does not produce any effect other than return value. Thus this call can be removed.

We are trying to reduce the false positives as much as possible, but in some cases this warning might be wrong. Common false-positive cases include:

- The method is designed to be overridden and produce a side effect in other projects which are out of the scope of the analysis.
- The method is called to trigger the class loading which may have a side effect.
- The method is called just to get some exception.

If you feel that our assumption is incorrect, you can use a `@CheckReturnValue` annotation to instruct SpotBugs that ignoring the return value of this method is acceptable.

SBSC_USE_STRINGBUFFER_CONCATENATION: Method concatenates strings using + in a loop

The method seems to be building a String using concatenation in a loop. In each iteration, the String is converted to a StringBuffer/StringBuilder, appended to, and converted back to a String. This can lead to a cost quadratic in the number of iterations, as the growing string is recopied in each iteration.

Better performance can be obtained by using a StringBuffer (or StringBuilder in Java 5) explicitly.

For example:

```
// This is bad
String s = "";
for (int i = 0; i < field.length; ++i) {
    s = s + field[i];
}
```

```
// This is better
StringBuffer buf = new StringBuffer();
for (int i = 0; i < field.length; ++i) {
    buf.append(field[i]);
}
String s = buf.toString();
```

SF_SWITCH_NO_DEFAULT: Switch statement found where default case is missing

This method contains a switch statement where default case is missing. Usually you need to provide a default case.

Because the analysis only looks at the generated bytecode, this warning can be incorrect triggered if the default case is at the end of the switch statement and the switch statement doesn't contain break statements for other cases.

SIC_INNER_SHOULD_BE_STATIC: Should be a static inner class

This class is an inner class, but does not use its embedded reference to the object which created it. This reference makes the instances of the class larger, and may keep the reference to the creator object alive longer than necessary. If possible, the class should be made static.

SS_SHOULD_BE_STATIC: Unread field: should this field be static?

This class contains an instance final field that is initialized to a compile-time static value. Consider making the field static.

ST_WRITE_TO_STATIC_FROM_INSTANCE_METHOD: Write to static field from instance method

This instance method writes to a static field. This is tricky to get correct if multiple instances are being manipulated, and generally bad practice.

UC_USELESS_OBJECT: Useless object created

Our analysis shows that this object is useless. It's created and modified, but its value never go outside of the method or produce any side-effect. Either there is a mistake and object was intended to be used or it can be removed.

This analysis rarely produces false-positives. Common false-positive cases include:

- This object used to implicitly throw some obscure exception.
- This object used as a stub to generalize the code.
- This object used to hold strong references to weak/soft-referenced objects.

URF_UNREAD_FIELD: Unread field

This field is never read. Consider removing it from the class.

URF_UNREAD_PUBLIC_OR_PROTECTED_FIELD: Unread public/protected field

This field is never read. The field is public or protected, so perhaps it is intended to be used with classes not seen as part of the analysis. If not, consider removing it from the class.

WMI_WRONG_MAP_ITERATOR: Inefficient use of keySet iterator instead of entrySet iterator

This method accesses the value of a Map entry, using a key that was retrieved from a keySet iterator. It is more efficient to use an iterator on the entrySet of the map, to avoid the Map.get(key) lookup.